

Observational Constraints on Quintessence Models using BAO Surveys and SNe Ia Pantheon+ Datasets

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Abstract

While the standard Λ CDM model attributes the universe’s accelerated expansion to a static cosmological constant (Λ), it suffers from profound fine-tuning and coincidence problems. Dynamic scalar fields, collectively known as Quintessence, offer an alternative theoretical framework where the dark energy density and its equation of state evolve over time. This paper examines two distinct Quintessence scenarios: the inverse power-law Ratra-Peebles model and the cosine-potential pseudo Nambu-Goldstone Boson (pNGB) model. We constrain the parameter spaces of these models by fitting theoretical distance moduli and volume-averaged distance ratios to high-redshift observational data from the Pantheon+ Type Ia Supernovae dataset and various Baryonic Acoustic Oscillation (BAO) surveys.

Keywords: *Quintessence, Cosmology, Dark Energy*

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1. INTRODUCTION

The discovery of late time accelerated expansion of the universe has been a challenge in modern theoretical physics. According to the standard Λ CDM model, this acceleration is attributed to an exotic form of energy with negative pressure called **dark energy**. In this concordance model, dark energy is mathematically represented by the cosmological constant (Λ), implying a constant energy density throughout cosmic history.

However this theory results in two major problems: **fine tuning problem** and **coincidence problem**.

The Fine-Tuning Problem

In modern cosmology, vacuum energy is the primary theoretical candidate proposed to explain the physical nature of dark energy. According to Quantum Field Theory, vacuum energy is the zero point energy of the quantum fields. However, theoretical calculations predict a vacuum energy density approximately 10^{120} times larger than the observed cosmological constant (Λ).

The Coincidence Problem

The coincidence problem is the coincident observation of the matter density and dark energy density being of the same order. The matter density evolves as being proportional to inverse cube of the scale factor while dark energy is constant with time.

These discrepancies have motivated the exploration of dynamic alternatives. Among the most prominent candidates is **Quintessence**, a dynamic time-evolving scalar field (ϕ) that provides a mechanism for a varying equation of state. Unlike the static nature of Λ , quintessence allows the dark energy density to evolve over time.

In this paper, we review the quintessence models governed by the Klein-Gordon field equation and the Friedmann equations. We specifically examine two distinct types of potentials: the inverse power-law potential (**Ratra-Peebles model**) and the cosine potential (**pseudo Nambu-Goldstone Boson or pNGB model**). To evaluate the feasibility of these models, we perform parameter fitting to determine the **Hubble parameter** (H) using the latest observational constraints. Our analysis utilizes high-redshift data from **Type Ia Supernovae (Pantheon+ dataset)** and **Baryon Acoustic Oscillations (BAO)** to compare the dynamic scalar field approach against the standard Λ CDM paradigm.

2. QUINTESSENCE

Quintessence is a scalar field, denoted by ϕ , which is dynamical in nature and is governed by the Klein-Gordon field equation

$$\ddot{\phi} + 3H\dot{\phi} + \frac{\partial V(\phi)}{\partial \phi} = 0 \quad (1)$$

Where H is the Hubble parameter, given by the Friedmann equation (taking $\hbar = c = 1$):

$$H^2 = \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi}{3}(\rho_m + \rho_r + \frac{1}{2}\dot{\phi}^2 + V(\phi)) \quad (2)$$

Where a is the cosmological scale factor, ρ_m is the matter density and ρ_r is the radiation energy density of the universe.

The equation of state of such a scalar field is given by

$$\omega_\phi = \frac{p}{\rho} = \frac{\frac{1}{2}\dot{\phi}^2 - V(\phi)}{\frac{1}{2}\dot{\phi}^2 + V(\phi)} \quad (3)$$

The behavior of the quintessence field depends on the form of the potential (V) taken for it. Here we will consider two such models, an inverse power law potential introduced by Ratra and Peebles [1] and a cosine potential introduced by Frieman et. al. [2], known as a pseudo Nambu-Goldstone Boson (pNGB) model.

2.1. Ratra-Peebles Model

Introduced by Bharat Ratra and P.J.E. Peebles in 1987 [1], this is one of the earliest quintessence models, and it assumes a potential of the form

$$V(\phi) = M^{4+\alpha}\phi^{-\alpha} \quad (4)$$

where M is of the dimensions of energy and $\alpha > 0$ is a positive number describing the steepness of the potential, constituting the two unknown parameters of this model.

The idea behind this model is that the field has stayed dynamic throughout the history of the universe, starting with $\omega \approx 0$, but only recently has it started 'slow-rolling' behavior, such that the kinetic energy is very small compared to the potential, thus approaching $\omega \rightarrow -1$ and 'freezing' due to Hubble friction. Thus this model is an example of a 'freezing model'. [4] [7]

The Klein-Gordon equation has a scaling solution given by: [5] [6]

$$\phi \propto a^{\frac{3(1+\omega_B)}{\alpha+2}}, \rho_\phi \propto a^{3(1+\omega_\phi)}, \omega_\phi = \frac{\alpha\omega_B - 2}{\alpha + 2} \quad (5)$$

Here, ω_B is the background equation of state parameter, which depends on the dominant component of the universe at any given time, such that in the radiation-dominated era, $\omega_B = \omega_r = \frac{1}{3}$, in the matter-domination era, $\omega_B = \omega_m = 0$.

In this way, we can see that the solution depends on and thus 'tracks' the background, which leads to such models being called 'tracker' models. [3]

The field in the radiation-domination era has the evolution:

$$\phi = \left(\frac{\alpha(2+\alpha)^2 M^{4+\alpha}}{6+\alpha} \right)^{\left(\frac{1}{2+\alpha}\right)} t^{\left(\frac{2}{2+\alpha}\right)} \quad (6)$$

And in the matter-domination era the field has the same time evolution behavior (varying as $t^{\left(\frac{2}{2+\alpha}\right)}$)

Importantly, when the energy density of matter reduces well below that of the quintessence field ($\rho_m < \rho_\phi$), we get the field evolution:

$$\phi = M \left(\frac{\alpha \left(2 + \frac{\alpha}{2}\right)}{\sqrt{24\pi G}} \right)^{\left(\frac{1}{2+\alpha}\right)} t^{\left(\frac{1}{2+\alpha}\right)} \quad (7)$$

This gives us

$$\ln a \propto t^{\frac{2}{2+\alpha}} \quad (8)$$

This evolution is close to but slower than that arrived at by using a cosmological constant, which gives $\ln a = t$

The tracker behavior also ensures that the solution is an attractor which is joined by the field from a wide range of the initial field energy density ($10^{-37} \text{ GeV}^4 < \rho_\phi < 10^{61} \text{ GeV}^4$). [5] Thus it solves the coincidence problem well.

This model however does not solve the fine-tuning problem, as it takes the fine tuning of Λ and shifts it to the fine tuning of the mass of this field ($\approx 10^{-33} \text{ eV}$). Later, various models were introduced to solve this new fine tuning problem, like the pNGB model discussed below.

2.2. Pseudo Nambu-Goldstone Bosons (pNGB) model

The pseudo-Nambu-Goldstone boson (pNGB) model, developed by Joshua A. Frieman et al. in 1995 [2], is motivated by symmetry principles in particle physics. It describes dark energy as a scalar field arising from the breaking of a global symmetry, leading to a naturally flat potential. It assumes a cosine form of an axion-like potential [8]

$$V(\phi) = M^4 \left[1 + \cos\left(\frac{\phi}{f}\right) \right] \quad (9)$$

Here M is the energy scale, and f is the symmetry breaking scale, which are the two parameters for this model which have to be fitted from the data.

This model is based on breaking of Shift Symmetry in the field of pure Goldstone Bosons, which have 0 mass. This symmetry breaking gives the field a very small non-zero mass and also protects it from blowing up from quantum fluctuations, which can explain the very small required mass of the Quintessence field, thus effectively solving the fine-tuning problem.

The idea of this model is that the field initially starts from somewhere near the top of the potential curve, and it frozen there due to hubble friction in the early universe. Later on, as the quintessence field dominates over matter, the field starts to 'thaw' and starts slowly rolling down the potential curve in recent times, leading to similar expansion behavior as expected from a cosmological constant. This is why this is an example of a 'thawing' model [7].

This 'near-hilltop' approximation gives us the potential:

$$V(\phi) \approx 2M^4 - \frac{M^4}{2f^2}(\delta\phi)^2 \quad (10)$$

Where $\delta\phi = \phi - \pi f$ gives the deviation of the field from the point of maximum potential.

While both of these models show the same behavior at the present time, both are fundamentally different and would be distinguishable with perfect observations, however, due to observational error bars being large enough to encapsulate the difference between the predictions made by these models, both these models currently give us results equally close to the observed behavior today, and which one is correct cannot be found.

3. METHODOLOGY

3.1. Observational Data

We use two independent cosmological probes: The *Pantheon+* and various Baryon Acoustic Oscillation (BAO) surveys (6dFGS, SDSS, BOSS, eBOSS).

The *Pantheon+* dataset consists of 1048 type Ia supernovae spanning a redshift range of $0 < z \lesssim 2.3$.

For BAO, we use a compilation of measurements covering redshifts $z = 0.106$ to $z = 2.33$, expressed in terms of the volume-averaged distance ratio $D_V(z)/r_s$, along with their corresponding observational errors. Due to the expansion of the Universe, the shape and radius of the BAO spheres differ from the sound horizon radius (r_s) observed in the CMBR.

We compare the **Distance Modulus** (μ) from the *Pantheon+* dataset and the **volume-averaged distance ratio** $D_V(z)/r_s$ from the BAO survey dataset with theoretical calculations to fit parameters in specific Quintessence models and constrain their parameter space using the methods described in the following subsections. Both of these values are affected by the expansion of the universe and hence are functions of redshift in a specific way.

3.2. Theoretical Framework

We try to solve H as a function of z from the Klein-Gordon equation of state of the Quintessence field,

$$\ddot{\phi} + 3H\dot{\phi} + \frac{dV}{d\phi} = 0, \quad (11)$$

The Friedmann equation is given by

$$H^2 = \frac{8\pi G}{3} \times \rho \quad (\text{Assuming Flat Universe})$$

$$\rho_c \equiv \frac{3H_0^2}{8\pi G} \quad (\text{Critical Density})$$

$$\Omega \equiv \frac{\rho}{\rho_c}; \Rightarrow \rho = \Omega\rho_c$$

$$\Rightarrow H^2 = \frac{8\pi G}{3}(\Omega\rho_c) = \frac{8\pi G}{3} \left[\Omega \left(\frac{3H_0^2}{8\pi G} \right) \right] = \Omega H_0^2$$

$$\Rightarrow \frac{H^2}{H_0^2} = \Omega$$

$$E(z) \equiv \frac{H}{H_0} = \sqrt{\Omega} \quad (\text{To reduce floating point precision})$$

$$\Rightarrow E(z) = \sqrt{\Omega_m + \Omega_\phi} \quad (\text{Assuming matter and field to be dominant})$$

(We have written $E(z)$ in the code as $H(z)$ and Ω as ρ ,

$$H(z) = \sqrt{\rho_m + \rho_\phi}, \quad (12)$$

, for ease of coding)

We vary the parameters of specific quintessence model and present day matter density Ω_{m0} . Here, the matter density evolves as $\Omega_m = \Omega_{m0}a^{-3}$ and the scalar field energy density is

$$\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V(\phi). \quad (13)$$

From solving the differential equation numerically, we get $H(z)$. Then, we find the co-moving distance as,

$$D_C(z) = \int_0^z \frac{dz'}{H(z')}. \quad (14)$$

The luminosity distance is then

$$D_L(z) = (1+z)D_C(z), \quad (15)$$

and the corresponding theoretical distance modulus is

$$\mu_{th} = 5 \log_{10} (D_L \text{ (in Mpc)}) + 25. \quad (16)$$

For BAO, we compute the volume-averaged distance

$$D_V(z) = \left[\frac{z D_C^2(z)}{H(z)} \right]^{1/3}, \quad (17)$$

and compare the ratio $D_V(z)/r_s$ with observations, where r_s is approximated as a constant sound horizon scale, where r_s is the sound horizon found in CMBR.

3.3. Numerical Implementation

Language : Python

Libraries : Numpy, Pandas, Scipy

The differential equation is solved numerically using the *solve_ivp* integrator from early times up to the present epoch in discrete time steps. In the *get_Hubble_function* function, we vary ϕ , $\dot{\phi}$ & a over time and compute value of H at each timestep and effectively evaluate $H(t)$. Initial conditions are specified at a small scale factor, and the evolution is terminated when $a = 1$. We find z at each timestep using the $a(t)$ array to get $z(t)$. The resulting $H(z)$ is interpolated using cubic

interpolation (using *interp1d* function) to allow efficient evaluation at arbitrary redshifts.

Distance measures are computed using numerical quadrature (*quad* function for integrations) for each redshift in the observational datasets. Theoretical predictions for both Supernovae and BAO observables are thus obtained for any given set of model parameters.

Model parameters (e.g. $\alpha, M, f, \Omega_{m0}$) are estimated by minimizing a combined chi-square function,

$$\chi^2 = \chi_{\text{SNe}}^2 + \chi_{\text{BAO}}^2, \quad (18)$$

where

$$\chi_{\text{SNe}}^2 = \sum_i \left(\frac{\mu_{\text{obs},i} - \mu_{\text{th},i}}{\sigma_i} \right)^2, \quad (19)$$

and

$$\chi_{\text{BAO}}^2 = \sum_i \left(\frac{(D_V/r_s)_{\text{obs},i} - (D_V/r_s)_{\text{th},i}}{\sigma_i} \right)^2. \quad (20)$$

The minimization is performed using the bounded $L - \text{BFGS} - B$ algorithm. The best-fit parameters correspond to the global minimum of χ^2 . where N is the total number of data points, and p is the number of free parameters.

Code

Here are the codes used for fitting:

1) Fitting Parameters of the Ratra-Peebles Model:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp, quad
from scipy.optimize import minimize
from scipy.interpolate import interp1d

#LOAD SNe DATA
url = 'https://raw.githubusercontent.com/dscolnic/Pantheon/master
↳ /lcp_param_full_long.txt'
sne_data = pd.read_csv(url, sep='s+')

sne_data = sne_data.sort_values(by='zcmb')
z_sne = sne_data['zcmb'].values
M_absolute = -19.36
mu_sne = sne_data['mb'].values - M_absolute
mu_error = sne_data['dmb'].values
print(f"Loaded {len(z_sne)} Supernovae.")

# LOAD BAO DATA (Extended Consensus: 6dFGS + SDSS MGS + BOSS +
↳ eBOSS)
z_bao = np.array([0.106, 0.15, 0.38, 0.51, 0.61, 0.70, 0.85,
↳ 1.52, 2.33])
dv_rs_data = np.array([2.98, 4.47, 10.23, 13.36, 15.51, 17.86,
↳ 19.50, 26.00, 31.22])
dv_rs_error = np.array([0.13, 0.16, 0.17, 0.21, 0.25, 0.33, 0.60,
↳ 1.15, 1.30])

# 3. The function to find H(z)

def get_Hubble_function(alpha, A, Omega_m0):
    #Solves the universe for V = A * phi^(-alpha)

    def V(phi): return A * (phi**(-alpha))
    def dV_dphi(phi): return -alpha * A * (phi**(-alpha - 1.0))

    def system(t, y):
        phi, phi_dot, a = y

        rho_m = Omega_m0 / (a**3)
        rho_phi = 0.5 * phi_dot**2 + V(phi)

        if rho_m + rho_phi <= 0: return [0, 0, 0]

        H = np.sqrt(rho_m + rho_phi)
```

```
dphi_dt = phi_dot
dphidot_dt = -3 * H * phi_dot - dV_dphi(phi)
da_dt = a * H

return [dphi_dt, dphidot_dt, da_dt]

def stop_today(t, y): return y[2] - 1.0
stop_today.terminal = True

y0 = [1e-3, 0.0, 0.001]

res = solve_ivp(system, [0, 100], y0, events=stop_today,
↳ max_step=0.05)

phi_arr, phidot_arr, a_arr = res.y
z_arr = (1.0 / a_arr) - 1.0

rho_m_arr = Omega_m0 / (a_arr**3)
rho_phi_arr = 0.5 * phidot_arr**2 + V(phi_arr)
H_arr = np.sqrt(rho_m_arr + rho_phi_arr)

H_interp = interp1d(z_arr[:-1], H_arr[:-1], kind='cubic',
↳ fill_value="extrapolate")
return H_interp

# DISTANCES FOR SNE AND BAO

def calculate_theory(z_array_sne, z_array_bao, alpha, A, ho,
↳ Omega_m0):
    H_func = get_Hubble_function(alpha, A, Omega_m0)
    #NOTE: z_array_sne and z_array_bao are fetched data z points
    # Calculate SNe Ia Observables (mu)
    mu_theory = []
    for z in z_array_sne:
        Dc, _ = quad(lambda x: 1.0 / H_func(x), 0, z)
        Dl = (1 + z) * Dc
        Dl_Mpc = Dl * (300000.0 / ho)
        mu = 5 * np.log10(Dl_Mpc) + 25
        mu_theory.append(mu)

    # Calculate BAO Observables (Dv/rs)
    bao_theory = []
    for z in z_array_bao:
        Dc, _ = quad(lambda x: 1.0 / H_func(x), 0, z)
        Dv = (z * (Dc**2) / H_func(z))**((1.0/3.0)

        rs_approx = 147.0 / (300000.0 / ho)
        bao_theory.append(Dv / rs_approx)

    return np.array(mu_theory), np.array(bao_theory)

#chisqr
def objective_chi2(params):
    alpha = params[0]
    A = params[1]
    Omega_m0 = params[2]
    ho = 70.0
    mu_th, bao_th = calculate_theory(z_sne, z_bao, alpha, A, ho,
↳ Omega_m0)

    chi2_sne = np.sum(((mu_sne - mu_th) / mu_error)**2)
    chi2_bao = np.sum(((dv_rs_data - bao_th) / dv_rs_error)**2)
    return chi2_sne + chi2_bao

print("Starting 3-Parameter Fit (This will take a minute due to
↳ 1048 SNe points)...")
initial_guess = [0.5, 0.7, 0.3]
bounds = [(-0.01, 4.0), (0.01, 3.5), (0.01, 0.5)]
result = minimize(objective_chi2, initial_guess, bounds=bounds,
↳ method='L-BFGS-B')
print("\n--- RESULTS ---")
best_alpha = result.x[0]
best_A = result.x[1]
best_Om = result.x[2]
fixed_ho = 70.0
min_chi2 = result.fun
best_M = best_A ** (1.0 / (4+best_alpha))

# Calculating Degrees of Freedom and Reduced Chi-Squared
num_data_points = len(z_sne) + len(z_bao)
num_parameters = 3
dof = num_data_points - num_parameters
reduced_chi2 = min_chi2 / dof
```

```

print(f"Best fit Alpha: {best_alpha:.6f}")
print(f"Best fit Amplitude (A): {best_M:.6f}")
print(f"Omega_m0: {best_0m:.4f}")
print(f"Total Chi2: {min_chi2:.2f}")
print(f"Degrees of Freedom: {dof}")
print(f"Reduced Chi2: {reduced_chi2:.7f}")
textstr = '\n'.join(
    r'$\alpha$=%.6f' % (best_alpha, ),
    r'$M$=%.6f' % (best_M, ),
    r'$\Omega_{m0}$=%.4f' % (best_0m, ),
    r'$\chi^2_{\nu}$=%.7f' % (reduced_chi2, ))
props = dict(boxstyle='round', facecolor='white', alpha=0.8,
    ↪ edgecolor='gray')
plt.rcParams.update({
    "font.size": 8,
    "axes.labelsize": 8,
    "axes.titlesize": 9,
    "legend.fontsize": 7,
    "xtick.labelsize": 7,
    "ytick.labelsize": 7
})
# --- PLOT 1: SNe ---
mu_plot, _ = calculate_theory(z_sne, [], best_alpha, best_A,
    ↪ fixed_h0, best_0m)

fig1, ax1 = plt.subplots(figsize=(3.4, 2.6))

ax1.scatter(z_sne, mu_sne, color='black', alpha=0.4, s=5, label='
    ↪ Pantheon')
ax1.plot(z_sne, mu_plot, 'r-', linewidth=1.2, label='Best-fit
    ↪ model')

ax1.set_xlabel('Redshift $z$')
ax1.set_ylabel('Distance modulus $\mu$')
ax1.set_title('Ratra-Peebles SNe Ia Fit')
ax1.legend(loc='upper left', frameon=False)
ax1.grid(True, alpha=0.3)

ax1.text(0.97, 0.03, textstr, transform=ax1.transAxes,
    fontsize=6, verticalalignment='bottom',
    horizontalalignment='right', bbox=dict(facecolor='white'
    ↪ , alpha=0.7, edgecolor='none'))

plt.tight_layout()
plt.savefig('SNeRatraPeebles.pdf', dpi=500)
plt.show()
# --- PLOT 2: BAO ---
_, bao_plot_th = calculate_theory([], z_bao, best_alpha, best_A,
    ↪ fixed_h0, best_0m)

fig2, ax2 = plt.subplots(figsize=(3.4, 2.6))

ax2.errorbar(z_bao, dv_rs_data, yerr=dv_rs_error,
    fmt='ko', markersize=4, label='BAO data')

ax2.plot(z_bao, bao_plot_th, 'rX', markersize=6, label='Best-fit
    ↪ model')

ax2.set_xlabel('Redshift $z$')
ax2.set_ylabel('$D_V / r_s$')
ax2.set_title('Ratra-Peebles BAO Fit')
ax2.legend(loc='upper left', frameon=False)
ax2.grid(True, alpha=0.3)

ax2.text(0.97, 0.03, textstr, transform=ax2.transAxes,
    fontsize=6, verticalalignment='bottom',
    horizontalalignment='right', bbox=dict(facecolor='white'
    ↪ , alpha=0.7, edgecolor='none'))
plt.tight_layout()
plt.savefig('BAORatraPeebles.pdf', dpi=500)
plt.show()

```

Code 1. Code to fit Ratra-Peebles Model parameter space.**2) Fitting Parameters of the pNGB Model:**

```

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp, quad, cumulative_simpson
from scipy.optimize import minimize
from scipy.interpolate import interp1d

```

```

#LOAD SNe DATA
url = 'https://raw.githubusercontent.com/dscolnic/Pantheon/master
    ↪ /lcpam_full_long.txt'
sne_data = pd.read_csv(url, sep='\s+')
sne_data = sne_data.sort_values(by='zcmb')
z_sne = sne_data['zcmb'].values
M_absolute = -19.36
mu_sne = sne_data['mb'].values - M_absolute
mu_error = sne_data['dmb'].values
print(f"Loaded {len(z_sne)} Supernovae.")

# LOAD BAO DATA (Extended Consensus: 6dFGS + SDSS MGS + BOSS +
    ↪ eBOSS)
z_bao = np.array([0.106, 0.15, 0.38, 0.51, 0.61, 0.70, 0.85,
    ↪ 1.52, 2.33])
dv_rs_data = np.array([2.98, 4.47, 10.23, 13.36, 15.51, 17.86,
    ↪ 19.50, 26.00, 31.22])
dv_rs_error = np.array([0.13, 0.16, 0.17, 0.21, 0.25, 0.33, 0.60,
    ↪ 1.15, 1.30])

# 3. THE ENGINE: RATRA-PEEBLES POTENTIAL
def get_Hubble_function(f, MU, Omega_m0):

    def V(phi): return (MU**4) * (1.0 + np.cos(phi / f))
    def dV_dphi(phi): return -( (MU**4) / f ) * np.sin(phi / f)

    def system(t, y):
        phi, phi_dot, a = y

        rho_m = Omega_m0 / (a**3)
        rho_phi = 0.5 * phi_dot**2 + V(phi)

        if rho_m + rho_phi <= 0: return [0, 0, 0]

        H = np.sqrt(rho_m + rho_phi)

        dphi_dt = phi_dot
        dphidot_dt = -3 * H * phi_dot - dV_dphi(phi)
        da_dt = a * H

        return [dphi_dt, dphidot_dt, da_dt]
    def stop_today(t, y): return y[2] - 1.0
    stop_today.terminal = True

    y0 = [0.2 * f, 0.0, 0.001]

    res = solve_ivp(system, [0, 1000], y0, events=stop_today,
        ↪ method='Radau', rtol=1e-4, atol=1e-6)

    if len(res.t_events[0]) == 0:
        raise ValueError("Universe did not reach a=1.0")

    phi_arr, phidot_arr, a_arr = res.y
    z_arr = (1.0 / a_arr) - 1.0
    rho_m_arr = Omega_m0 / (a_arr**3)
    rho_phi_arr = 0.5 * phidot_arr**2 + V(phi_arr)
    H_arr = np.sqrt(rho_m_arr + rho_phi_arr)
    # Reverse arrays
    z_rev = z_arr[::-1]
    H_rev = H_arr[::-1]

    # Remove duplicates in z so that no repeated z values are
    ↪ there and Interpid doesnt crash
    z_unique, indices = np.unique(z_rev, return_index=True)
    H_unique = H_rev[indices]

    # Interpolate
    H_interp = interp1d(z_unique, H_unique, kind='cubic',
        ↪ fill_value="extrapolate")
    return H_interp

# DISTANCES FOR SNE AND BAO

def calculate_theory(z_array_sne, z_array_bao, f, MU, h0,
    ↪ Omega_m0):
    H_func = get_Hubble_function(f, MU, Omega_m0)

    z_sne_arr = np.array(z_array_sne)
    z_bao_arr = np.array(z_array_bao)
    # 1. Determine the maximum z needed for the grid (handle

```

```

    ↪ empty arrays for plotting)
if z_sne_arr.size > 0 and z_bao_arr.size > 0:
    z_max = max(np.max(z_sne_arr), np.max(z_bao_arr)) + 0.1
elif z_sne_arr.size > 0:
    z_max = np.max(z_sne_arr) + 0.1
elif z_bao_arr.size > 0:
    z_max = np.max(z_bao_arr) + 0.1
else:
    return np.array([]), np.array([])

# 2. Creating dense grid and evaluate H(z)
# Using 1001 points (odd number required for Simpsons rule)
z_grid = np.linspace(0, z_max, 1001)
H_grid = H_func(z_grid)

# 3. Integrate ONCE
Dc_grid = cumulative_simpson(1.0 / H_grid, x=z_grid, initial
    ↪ =0)

# 4. Create interpolator
Dc_func = interp1d(z_grid, Dc_grid, kind='cubic', fill_value=
    ↪ "extrapolate")

# Calculation for SNe Ia
if z_sne_arr.size > 0:
    Dc_sne = Dc_func(z_sne_arr)
    DL_sne = (1 + z_sne_arr) * Dc_sne
    DL_Mpc = DL_sne * (300000.0 / h0)
    mu_theory = 5 * np.log10(DL_Mpc) + 25
else:
    mu_theory = np.array([])

# Calculation for BAO
if z_bao_arr.size > 0:
    Dc_bao = Dc_func(z_bao_arr)
    H_bao = H_func(z_bao_arr)
    Dv_bao = (z_bao_arr * (Dc_bao**2) / H_bao)**(1.0/3.0)

    rs_approx = 147.0 / (300000.0 / h0)
    bao_theory = Dv_bao / rs_approx
else:
    bao_theory = np.array([])

return mu_theory, bao_theory

#Chisqr
def objective_chi2(params):
    f = params[0]
    MU = params[1]
    Omega_m0 = params[2]
    h0 = 70.0
    try:
        mu_th, bao_th = calculate_theory(z_sne, z_bao, f, MU, h0,
            ↪ Omega_m0)

        chi2_sne = np.sum(((mu_sne - mu_th) / mu_error)**2)
        chi2_bao = np.sum(((dv_rs_data - bao_th) / dv_rs_error)
            ↪ **2)

        return chi2_sne + chi2_bao
    except Exception:
        # If the parameter guess causes ODE failure or math
        ↪ errors,
        # return a high penalty to push L-BFGS-B away.
        return 1e10

print("\n\nStarting 3-Parameter Fit\n")
initial_guess = [2.0, 0.9, 0.3]
bounds = [(0.01, 4.0), (0.01, 5.0), (0.2, 0.5)]

result = minimize(objective_chi2, initial_guess, bounds=bounds,
    ↪ method='L-BFGS-B')

print("\n--- RESULTS ---")
best_f = result.x[0]
best_MU = result.x[1]
best_Om = result.x[2]
fixed_h0 = 70.0
min_chi2 = result.fun
# Calculate Degrees of Freedom and Reduced Chi-Squared
num_data_points = len(z_sne) + len(z_bao)
num_parameters = 3

```

```

dof = num_data_points - num_parameters
reduced_chi2 = min_chi2 / dof

print(f"Best fit f: {best_f:.6f}")
print(f"Best fit MU: {best_MU:.6f}")
print(f"Omega_m0: {best_Om:.4f}")
print(f"Total Chi2: {min_chi2:.2f}")
print(f"Degrees of Freedom: {dof}")
print(f"Reduced Chi2: {reduced_chi2:.7f}")

textstr = '\n'.join((
    r'$f$=%.7f$' % (best_f, ),
    r'$M$=%.7f$' % (best_MU, ),
    r'$\Omega_{m0}$=%.4f$' % (best_Om, ),
    r'$\chi^2_{\nu}$=%.7f$' % (reduced_chi2, )))
props = dict(boxstyle='round', facecolor='white', alpha=0.8,
    ↪ edgecolor='gray')
plt.rcParams.update({
    "font.size": 8,
    "axes.labelsize": 8,
    "axes.titlesize": 9,
    "legend.fontsize": 7,
    "xtick.labelsize": 7,
    "ytick.labelsize": 7
})

# --- PLOT 1: SNe ---
mu_plot, _ = calculate_theory(z_sne, [], best_f, best_MU,
    ↪ fixed_h0, best_Om)

fig1, ax1 = plt.subplots(figsize=(3.4, 2.6))

ax1.scatter(z_sne, mu_sne, color='black', alpha=0.4, s=5, label='
    ↪ Pantheon')
ax1.plot(z_sne, mu_plot, 'b-', linewidth=1.2, label='Best-fit
    ↪ model')

ax1.set_xlabel('Redshift $z$')
ax1.set_ylabel('Distance modulus $\mu$')
ax1.set_title('PNGS SNe Ia Fit')
ax1.legend(loc='upper left', frameon=False)
ax1.grid(True, alpha=0.3)

ax1.text(0.97, 0.03, textstr, transform=ax1.transAxes,
    fontsize=6, verticalalignment='bottom',
    horizontalalignment='right', bbox=dict(facecolor='white'
    ↪ , alpha=0.7, edgecolor='none'))
plt.tight_layout()
plt.savefig('SNePNGB.pdf', dpi=500)
# plt.savefig('SNe_Fit_RatraPeebles.png', dpi=500)
plt.show()

# --- PLOT 2: BAO ---
_, bao_plot_th = calculate_theory([], z_bao, best_f, best_MU,
    ↪ fixed_h0, best_Om)

fig2, ax2 = plt.subplots(figsize=(3.4, 2.6))

ax2.errorbar(z_bao, dv_rs_data, yerr=dv_rs_error,
    fmt='ko', markersize=4, label='BAO data')

ax2.plot(z_bao, bao_plot_th, 'bX', markersize=6, label='Best-fit
    ↪ model')
ax2.set_xlabel('Redshift $z$')
ax2.set_ylabel('$D_V / r_s$')
ax2.set_title('PNGS BAO Fit')
ax2.legend(loc='upper left', frameon=False)
ax2.grid(True, alpha=0.3)

ax2.text(0.97, 0.03, textstr, transform=ax2.transAxes,
    fontsize=6, verticalalignment='bottom',
    horizontalalignment='right', bbox=dict(facecolor='white'
    ↪ , alpha=0.7, edgecolor='none'))
plt.tight_layout()
plt.savefig('BAOPNGB.pdf', dpi=500)
# plt.savefig('BAO_Fit_RatraPeebles.png', dpi=500)
plt.show()

```

Code 2. Code to fit pNGS Model parameter space.

4. RESULTS

These are the fitted parameters and plots we got by applying the potentials in (4) (Ratra-Peebles) and (9) (pNGB):

4.1. Ratra-Peebles:

$$\begin{aligned}\alpha &= 0.000257 \\ M &= 0.930121 \quad (\text{in units of } meV) \\ \Omega_{m0} &= 0.2543 \\ \chi^2_\nu &= 1.0388087\end{aligned}$$

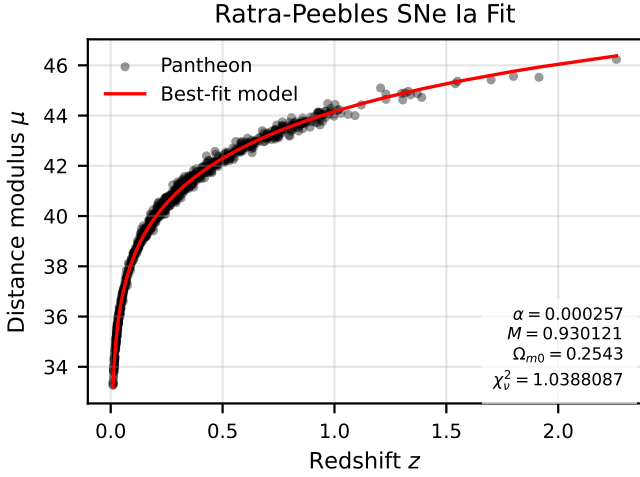


Figure 1. Distance Modulus vs Redshift from Survey Data and Functions Obtained Using Fitted Parameters of the Ratra-Peebles Model.

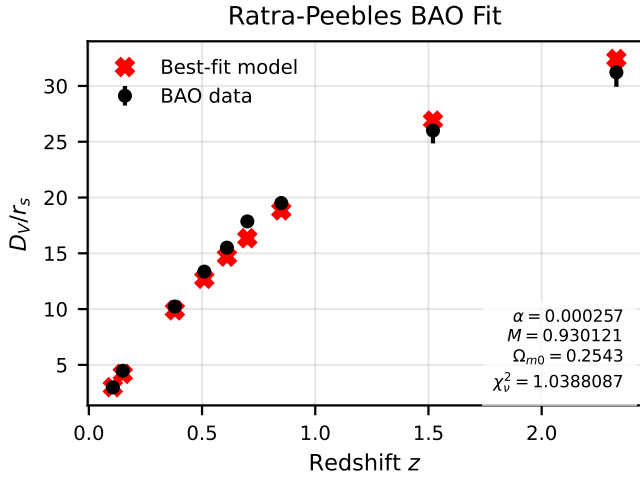


Figure 2. Volume-averaged Distance Ratio vs Redshift from Survey Data and Functions Obtained Using Fitted Parameters of the Ratra-Peebles Model.

4.2. Pseudo Nambu-Goldstone Bosons:

$$\begin{aligned}f &= 1.7340619 \quad (\text{in units of } M_{pl}) \\ M &= 0.7847793 \quad (\text{in units of } meV) \\ \Omega_{m0} &= 0.2535 \\ \chi^2_\nu &= 1.0387813\end{aligned}$$

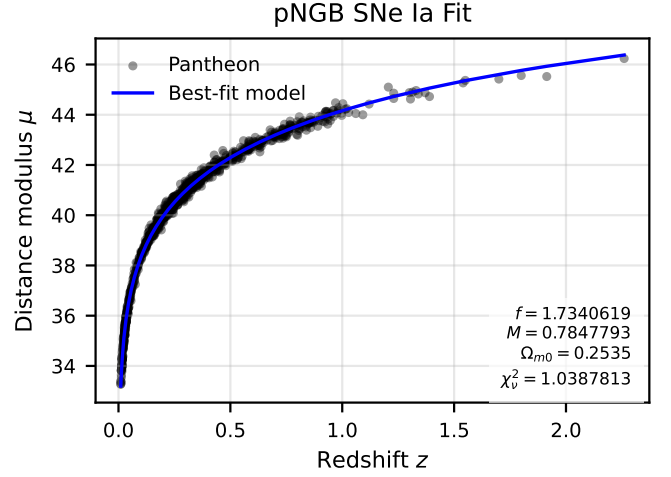


Figure 3. Distance Modulus vs Redshift from Survey Data and Functions Obtained Using Fitted Parameters of the Pseudo Nambu-Goldstone Bosons Model.

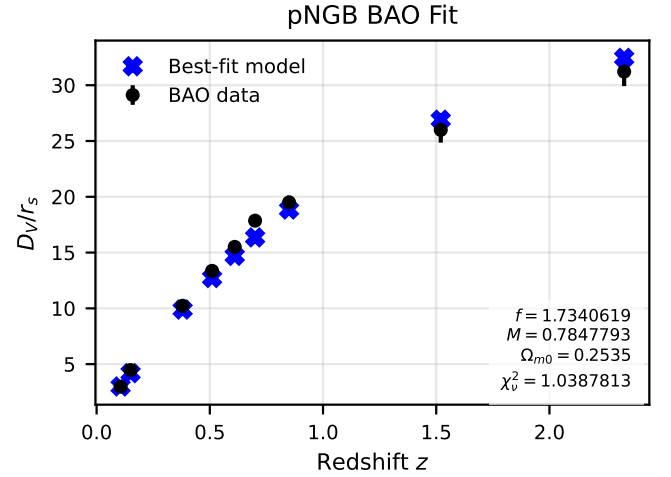


Figure 4. Volume-averaged Distance Ratio vs Redshift from Survey Data and Functions Obtained Using Fitted Parameters of the Pseudo Nambu-Goldstone Bosons Model.

Brief Note on Dimensions

In the discussion of Quintessence or generally in cosmology we use the natural unit where $\hbar = c = 1$. In this system, action $S = \int d^4x \mathcal{L}$ is dimensionless, meaning the Lagrangian density $\mathcal{L}$ must have dimensions of $[Mass]^4$ or $[Energy]^4$.

α (Ratra-Peebles Exponent): Dimensionless; Unit: None.

ϕ (Scalar Field): Dimension: $[Mass]^1$; Unit: M_{pl} .

$V(\phi)$ (Potential): Dimension: $[Mass]^4$; Unit: eV^4 .

M (Ratra-Peebles Mass Scale): Dimension: $[Mass]^1$; Unit: meV .

f (PGB Decay Constant): Dimension: $[Mass]^1$; Unit: M_{pl} .

M (PGB Amplitude Mass Scale): Dimension: $[Mass]^1$; Unit: meV .

5. CONCLUSION

1) The low fitted value of α in the Ratra-Peebles model mathematically flattens the inverse power-law potential ($V \approx M^4$). This prevents the scalar field from dynamically rolling, driving its equation of state toward $w \rightarrow -1$. This signifies that current high-redshift data overwhelmingly favors a static Cosmological Constant over dynamical dark energy. Due to this small exponent the field is said to be "slow rolling".

2) In the pNGB model, the large fitted value of f and extremely small initial condition on ϕ make the argument in cosine of the potential nearly zero. Hence, effectively the field sits on top of the potential and the potential acts nearly constant which again drives the equation of state parameter towards $w \rightarrow -1$.

3) We see the fitted value of f in pNGB model varies extremely with initial condition to the minimizer. f exhibits this degeneracy because in early-universe Hubble friction freezes the scalar field at the nearly flat peak of its potential. Since the field rolls extremely slowly, the universe's background expansion history becomes completely blind to the potential's width, preventing the data from constraining f to a unique value.

4) The reduced chi-squared (χ^2_ν) values for both the Ratra-Peebles and pNGB models are nearly identical; both are simply tending to the Λ CDM model. Current SNe Ia and BAO datasets are highly consistent with a standard $w = -1$ cosmology. Because both models possess mathematical limits that allow them to perfectly mimic a cosmological constant, the minimizer simply finds this exact same Λ CDM limit in both parameter spaces.

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